Enclustra User Schematics

Mercury PE1 Revision 4.6

Disclaimer

User schematics do not include proprietary Enclustra design elements, such as power supply circuits.
These circuits use optimized designs which are extremely compact and remain proprietary to Enclustra.
The user manual provides all necessary information to use the module and its interfaces.

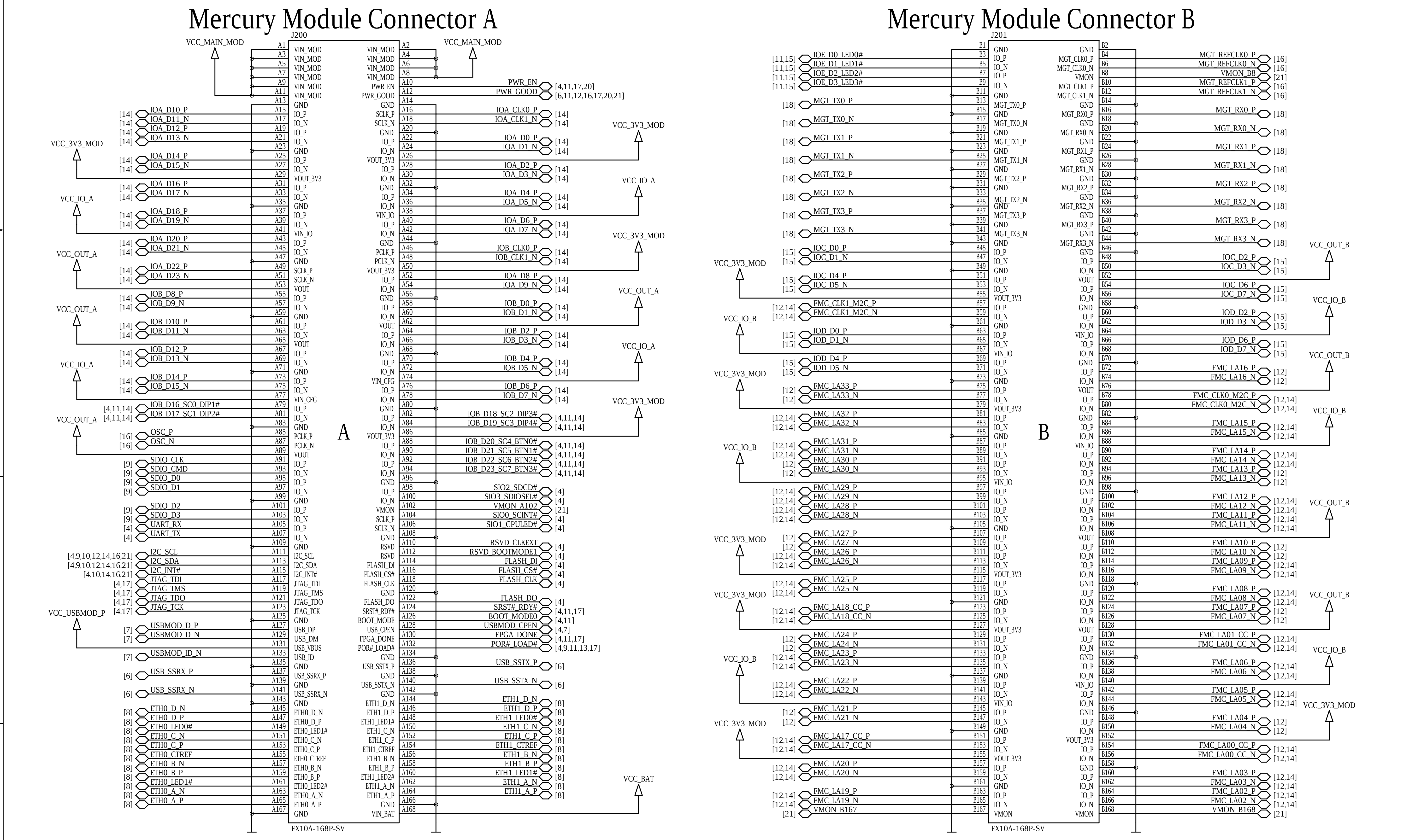
User schematics are provided after purchase of Enclustra hardware, and may also be provided in certain cases before purchase, to assist in product evaluation.
No part of the schematics may be copied or modified without written permission from Enclustra.

Full schematics, including a full bill-of-materials, may be available through the purchase of a hardware licence for the product in question.
Please contact Enclustra Sales for more information.

Note: DNE = Do Not Equip (parts not equipped by default)

Mercury PE1 Base Board

Revision 4.6



ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	2_MERCURY_MODULE_CONNECTORS_A/B	Customer No	0000	Revision	R4.6	DNE = do not equip
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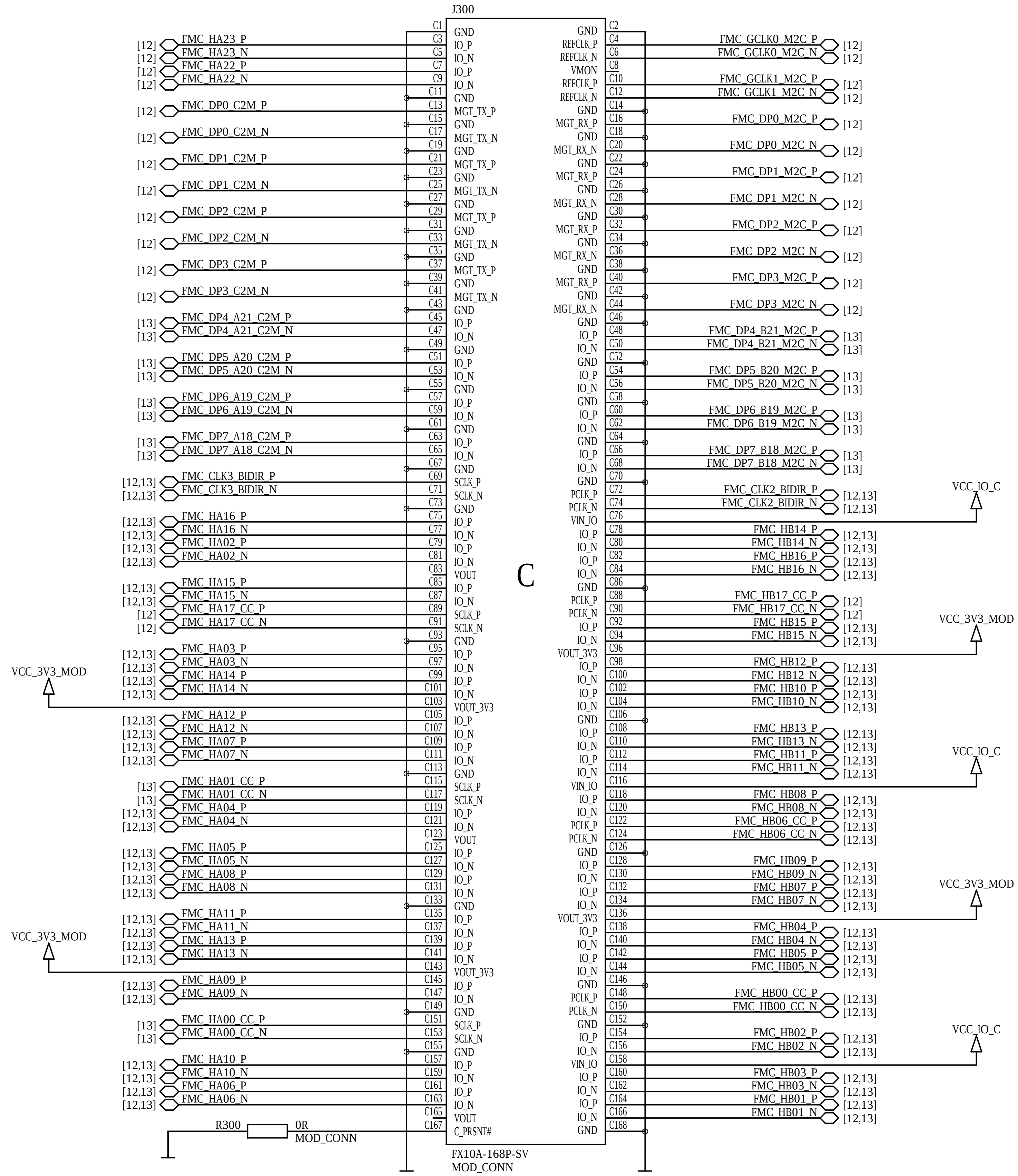
Mercury Module Connector C



ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	3_MERCURY_MODULE_CONNECTOR_C	Customer No	0000	Revision	R4.6	DNE = do not equip	Confidential
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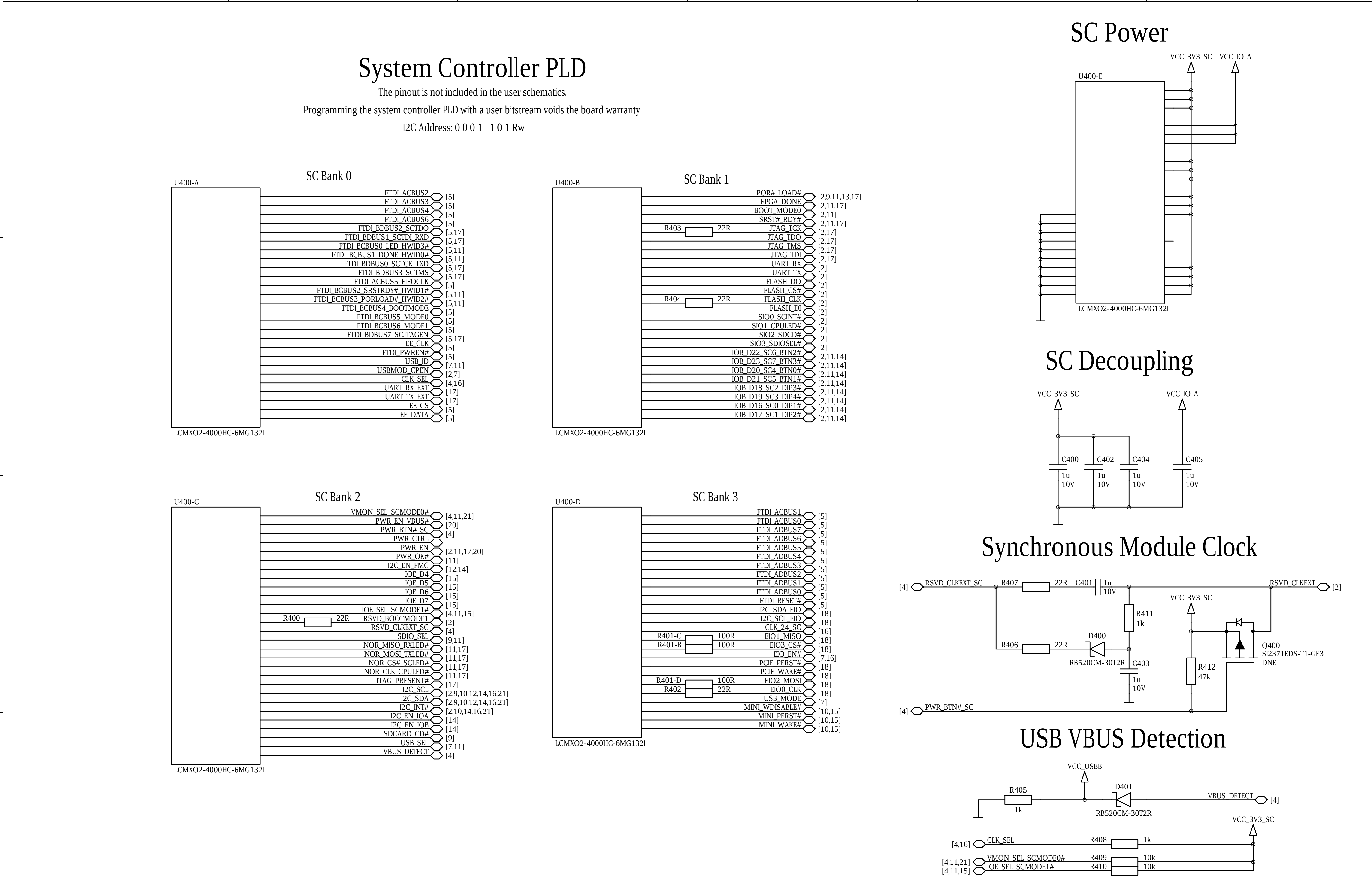

ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	4_SYSTEM_CONTROLLER_PLD	Customer No	0000	Revision	R4.6	DNE = do not equip	Confidential
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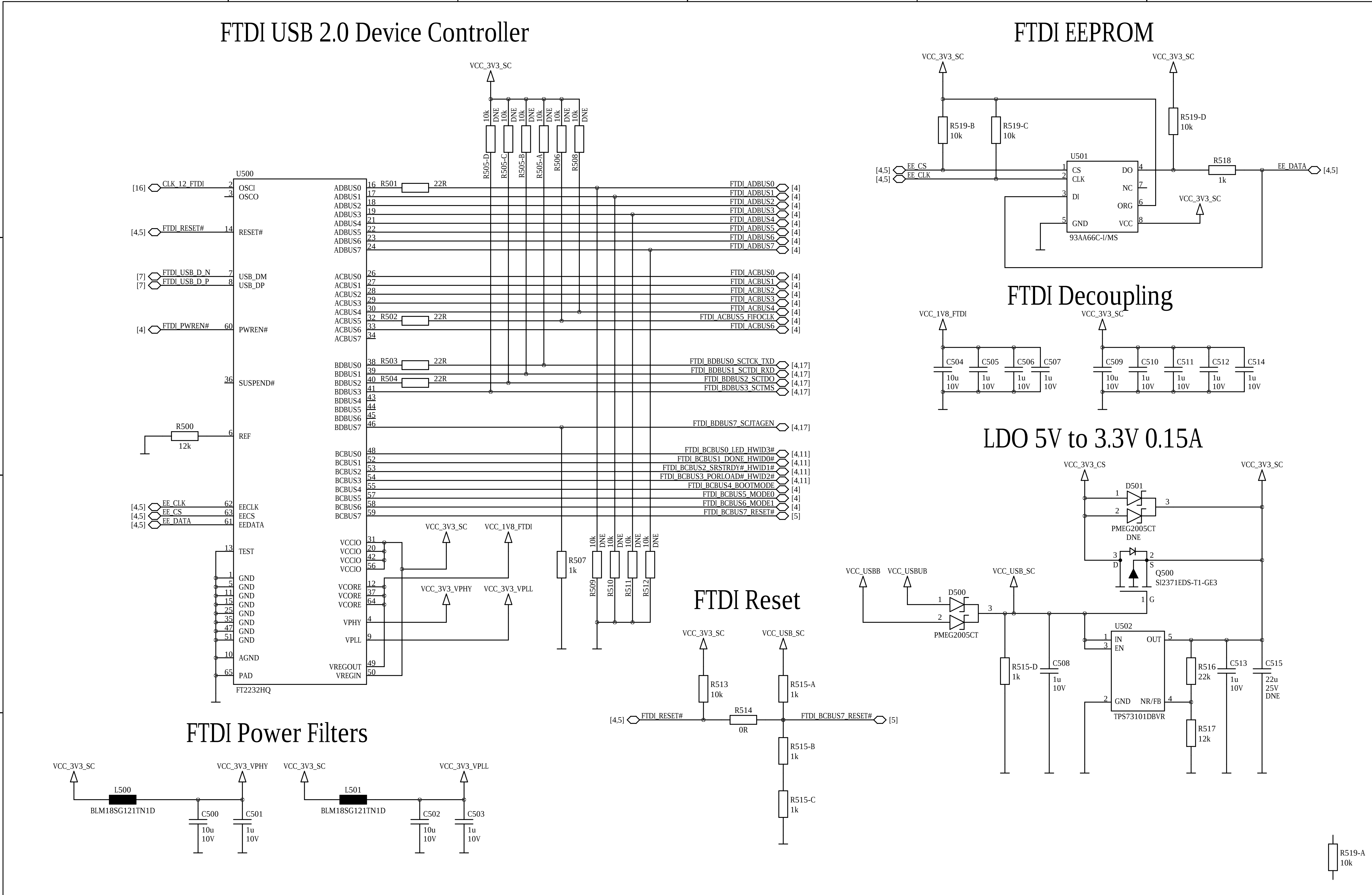

ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	5_USB_CONTROLLER	Customer No 0000	Revision R4.6	DNE = do not equip	Confidential
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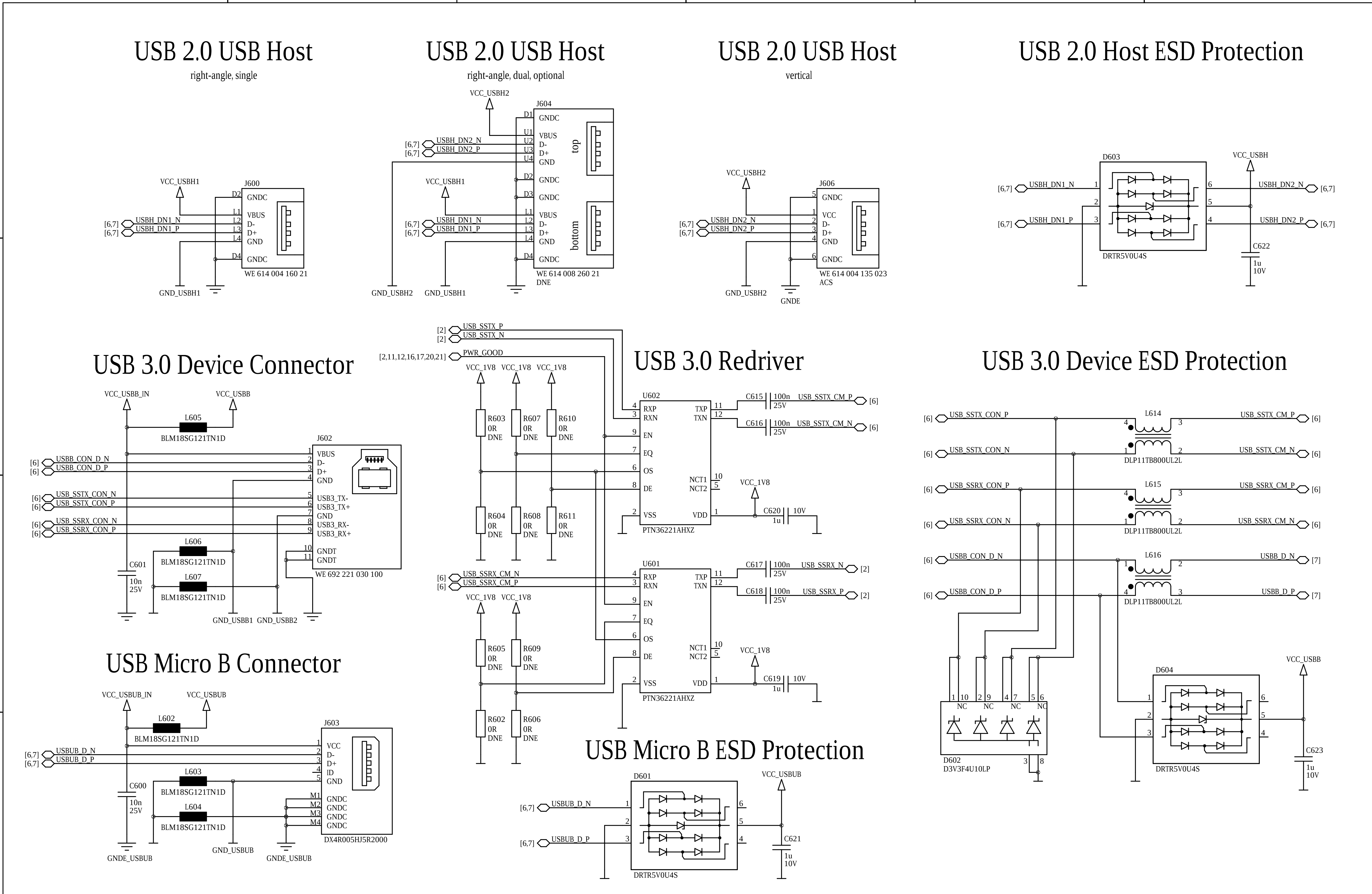

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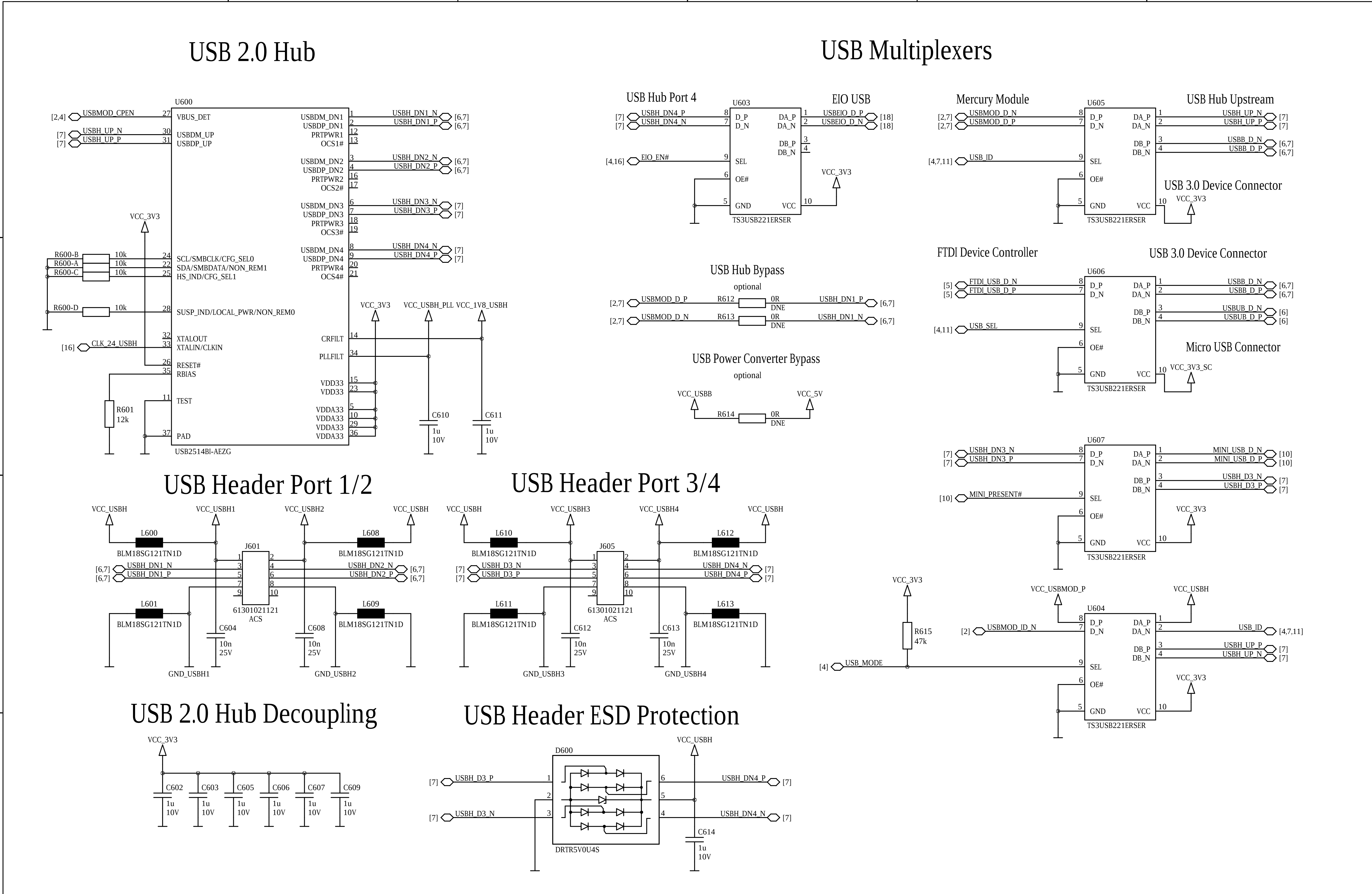

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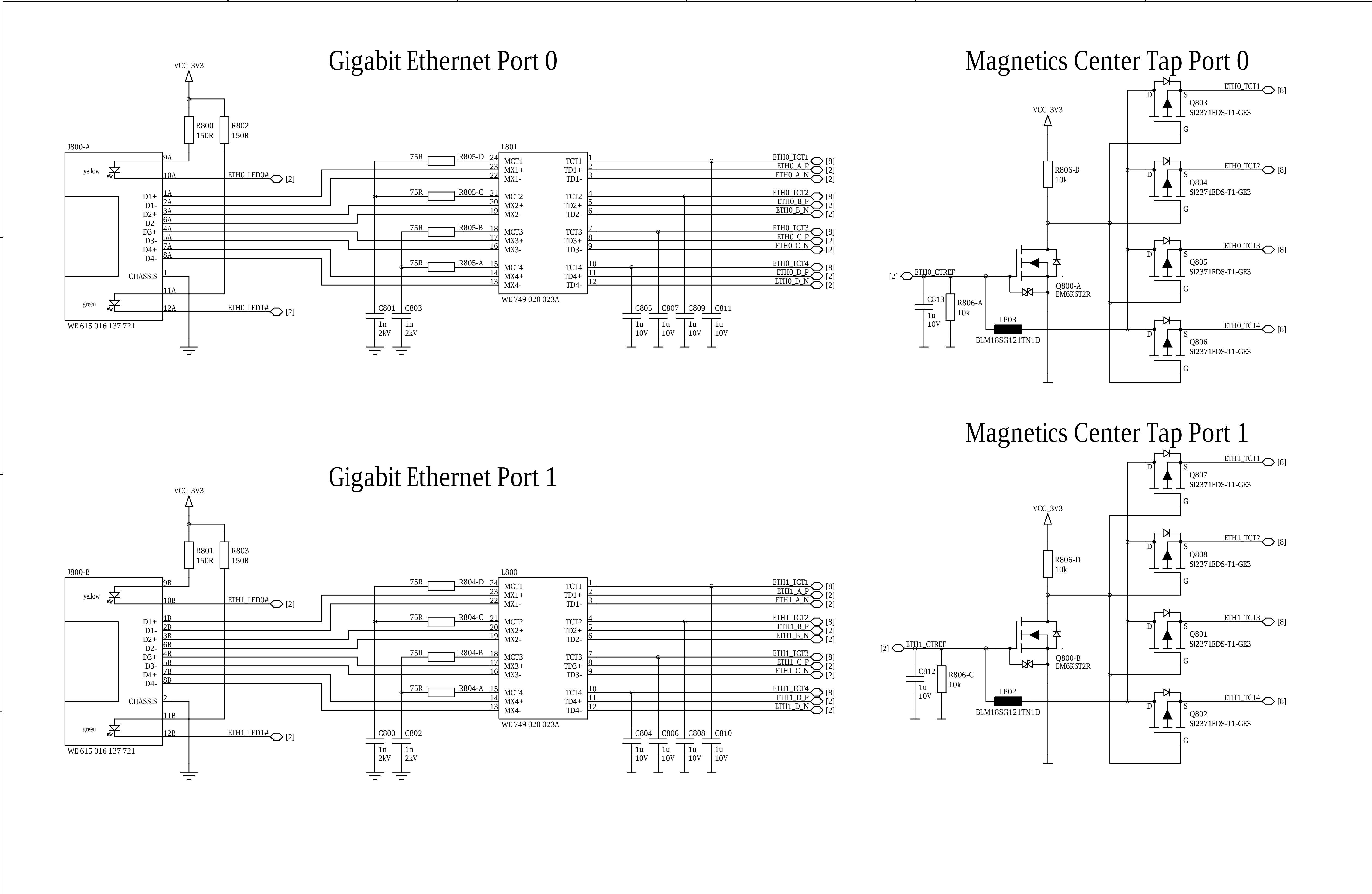

ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	8_DUAL_GIGABIT_ETHERNET	Customer No 0000	Revision R4.6	DNE = do not equip	Confidential
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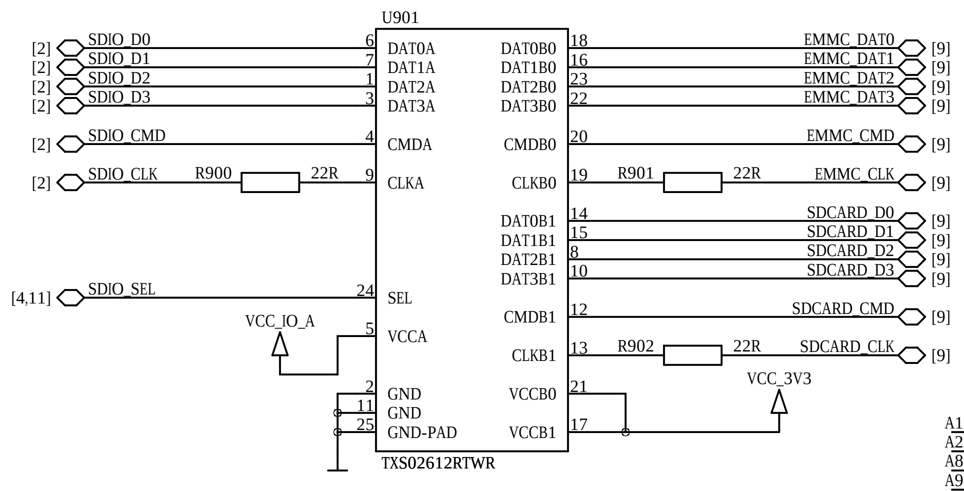
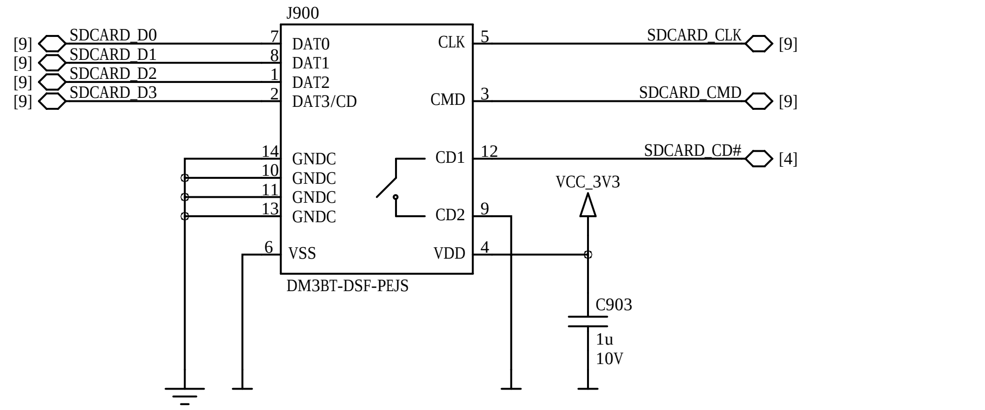
SDIO Level Shifter

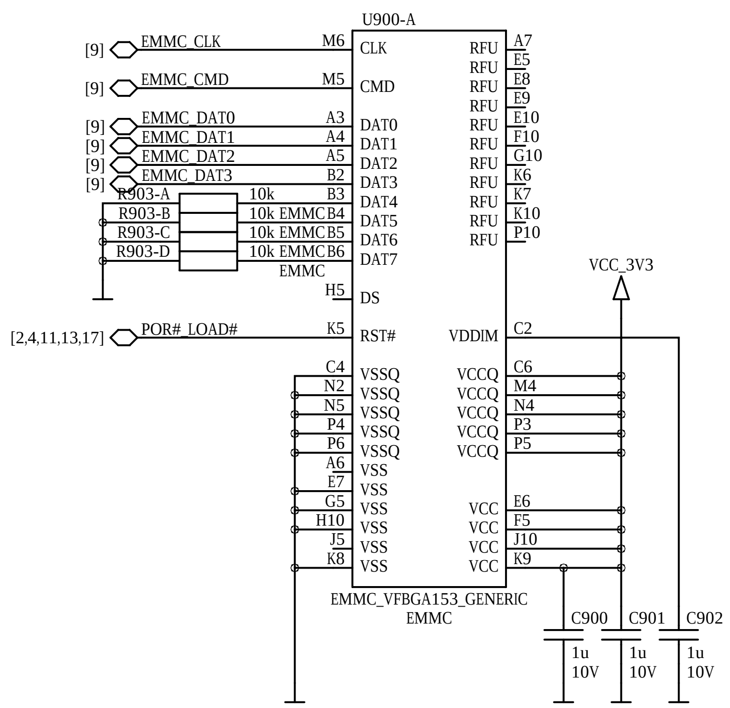


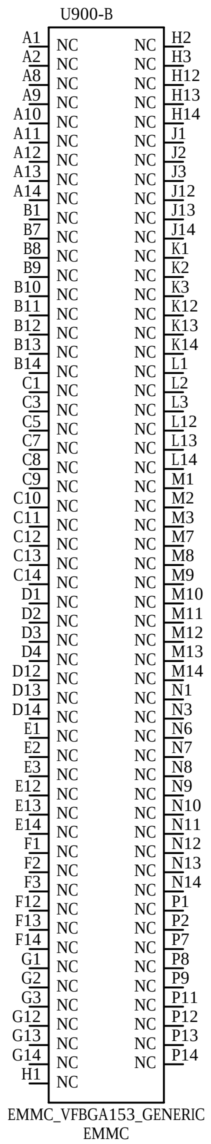
Micro SD-Card Slot



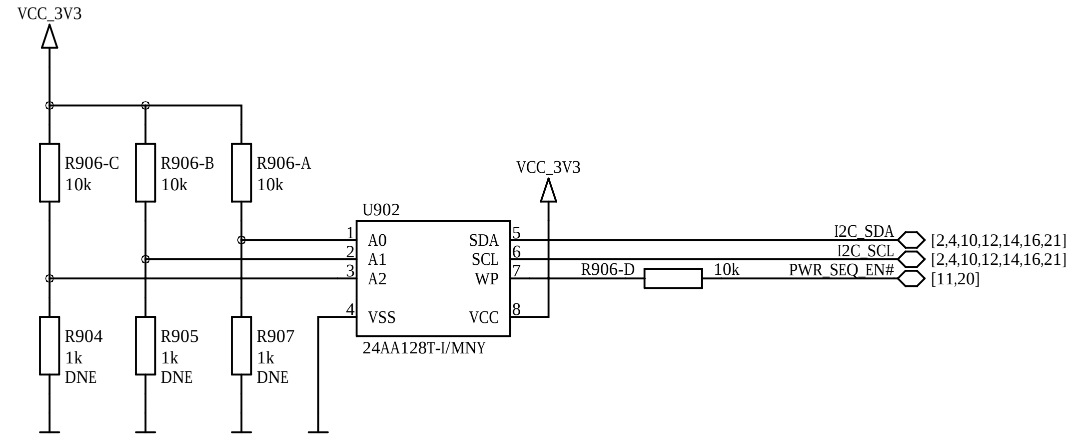
eMMC Managed NAND Flash

optional in some configurations





User EEPROM



I2C Address: 1 0 1 0 1 1 1 Rw

ENCLUSTRA	Copyright © 2020 by Enclustra GmbH	Sheet Name 9_SDCARD_EEPROM_EMMC	Customer No 0000	Revision R4.6	DNE = do not equip	Confidential
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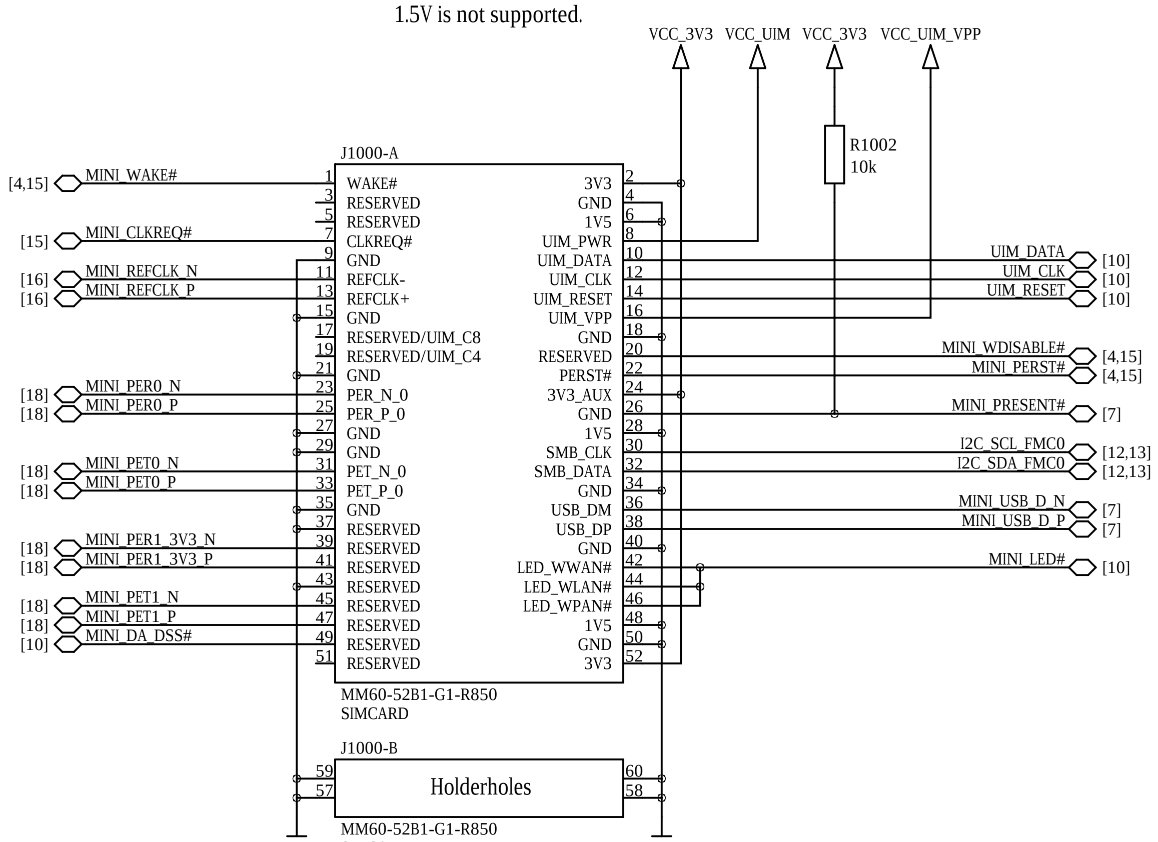
PCIe Mini / Mini SATA Card Holder

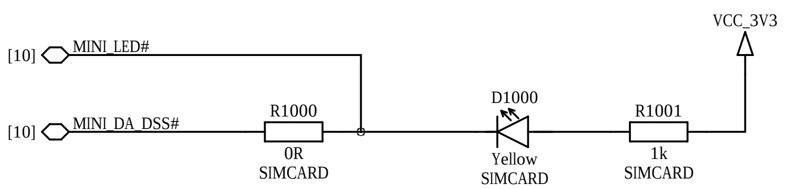
optional in some configurations

By default USB only. PCIe and SATA optional.

1.5V is not supported.

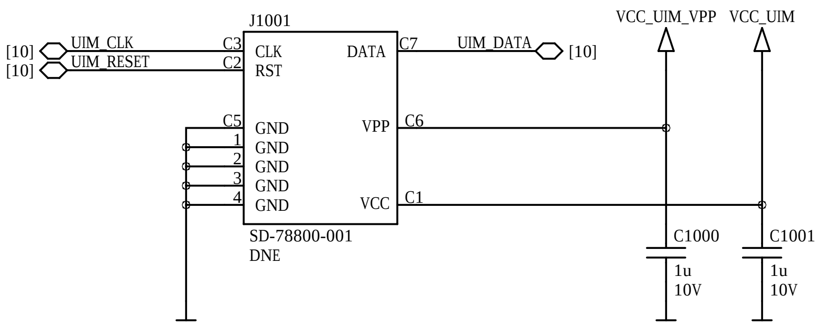


Mini Card Activity LED



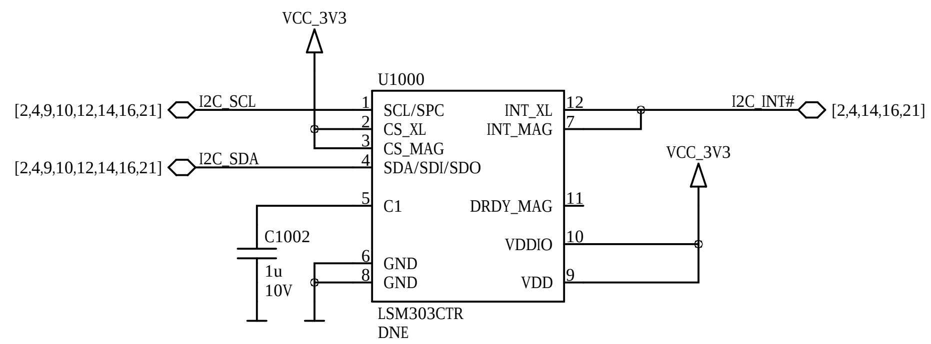
SIM Smart Card Slot

optional



Accelerometer / Magnetometer / Temperature Sensor

optional in some configurations



I2C Addresses

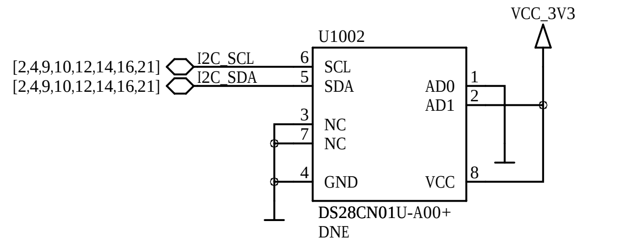
Accelerometer: 0 0 1 1 1 0 1 Rw

Magnetometer: 0 0 1 1 1 1 0 Rw

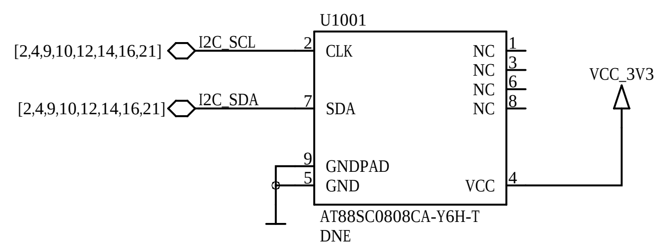
Secure EEPROMs

optional

for Enclustra use only



I2C Address: 1 0 1 0 1 0 0 Rw



I2C Address: 0 0 0 1 X 1 1 Rw

ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	10_MINI_CARD_SENSOR	Customer No	0000	Revision	R4.6	DNE = do not equip	Confidential
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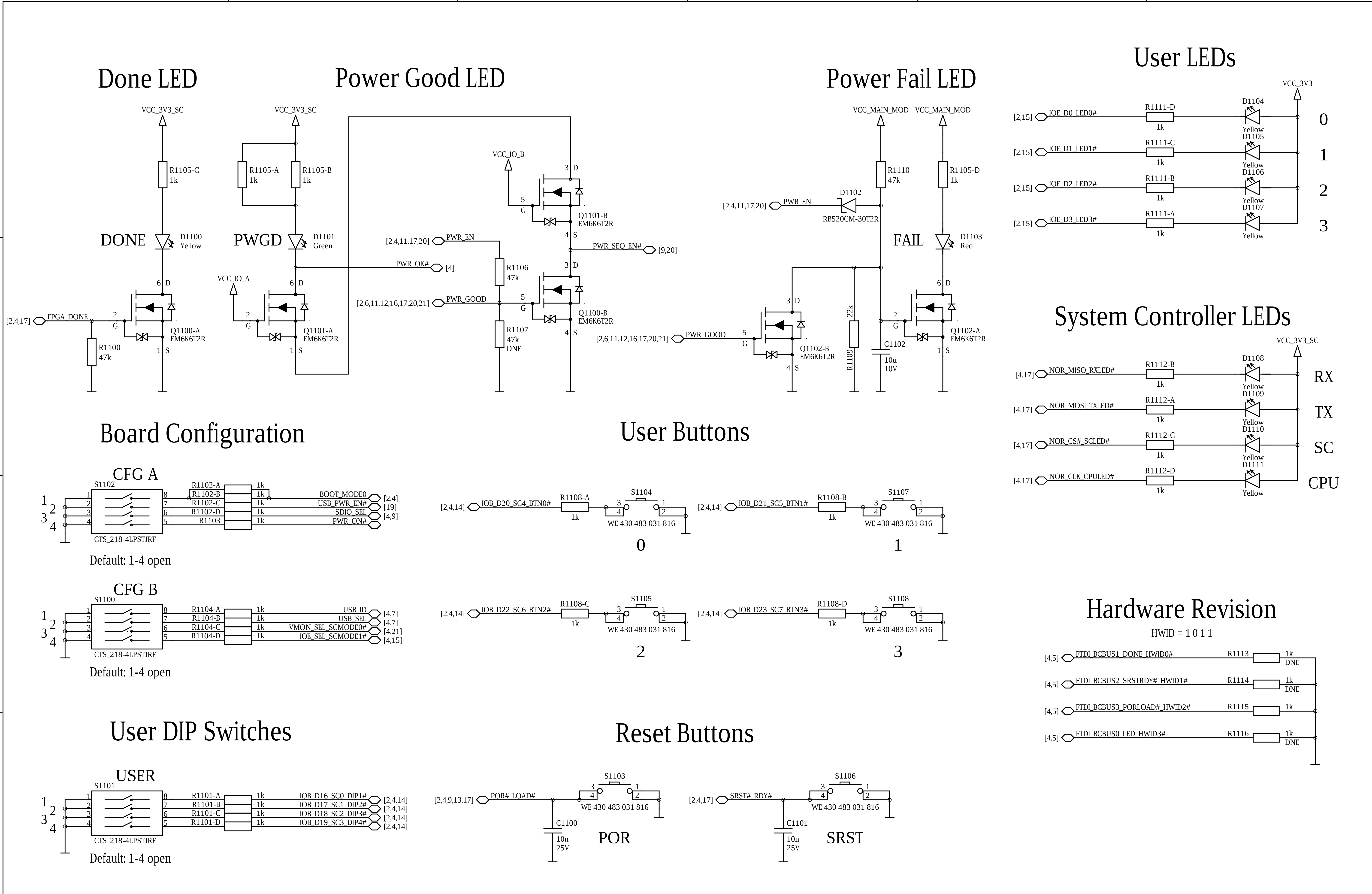

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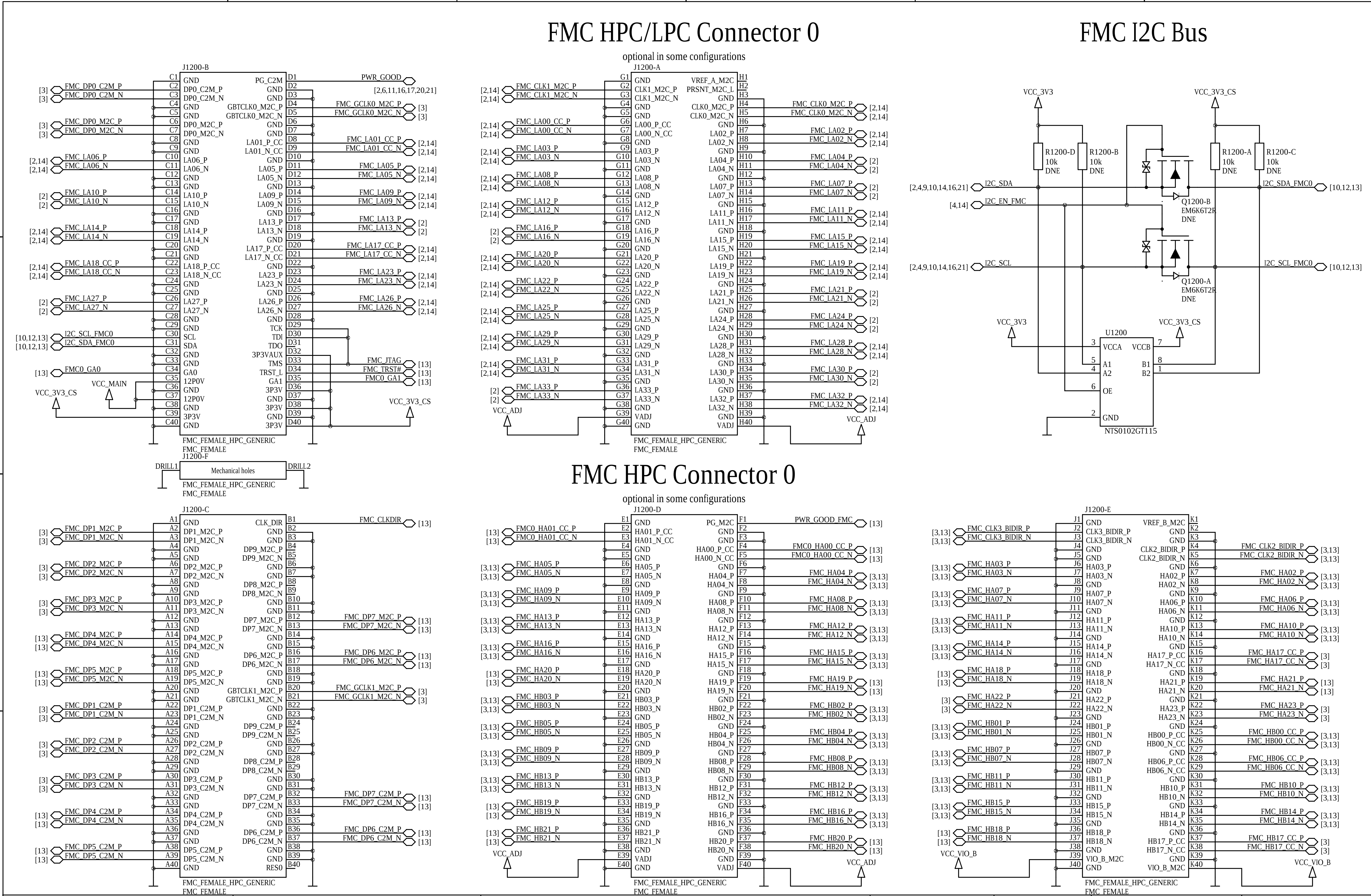

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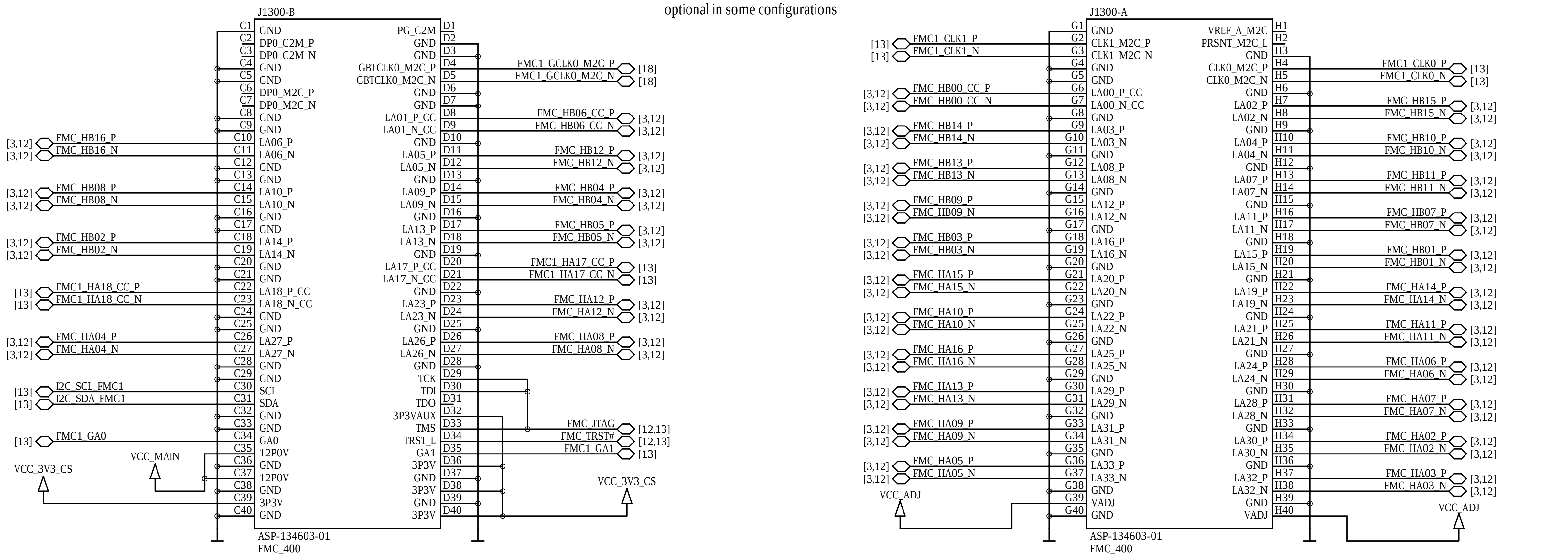
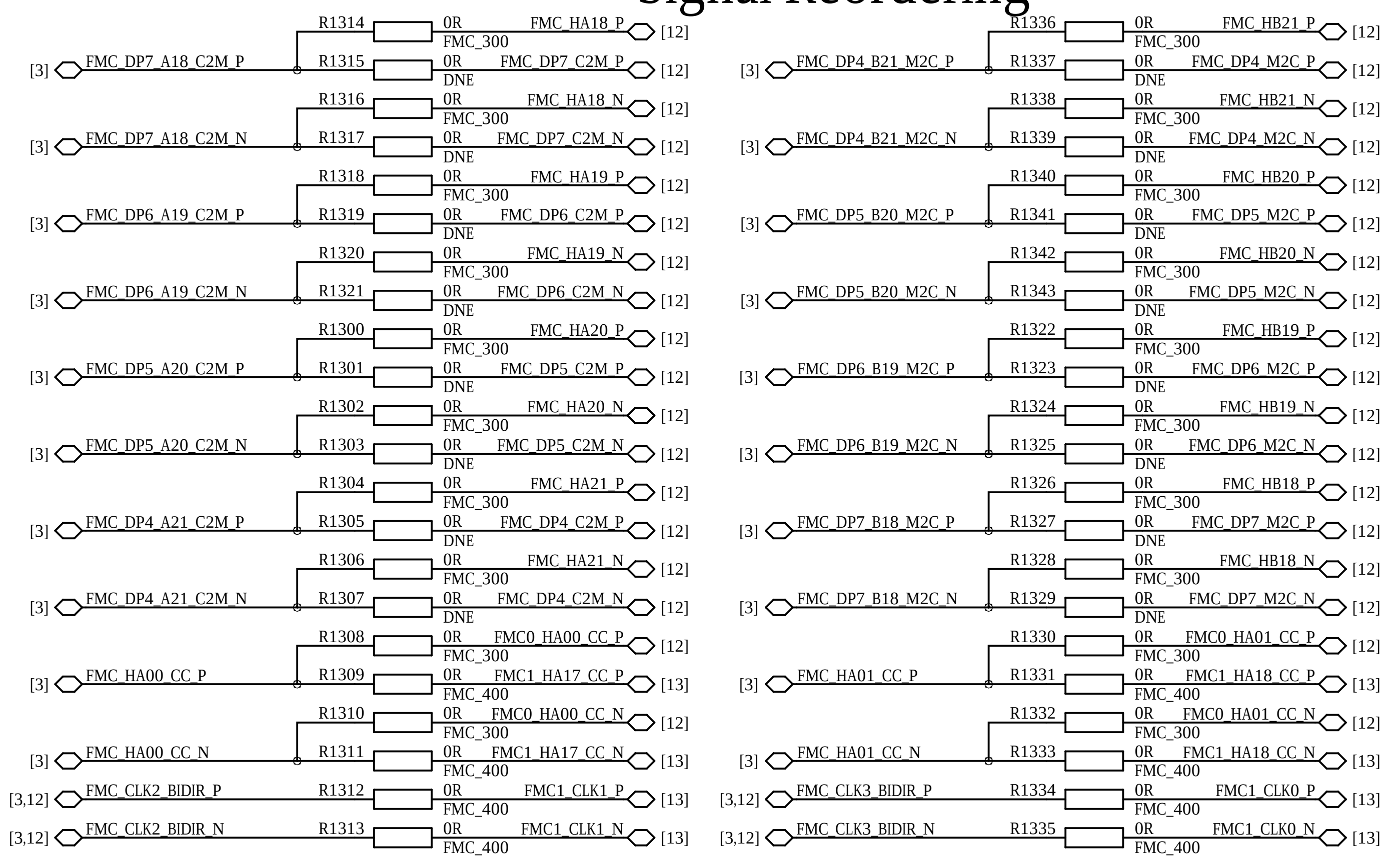
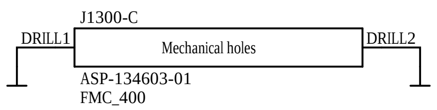
FMC LPC Connector 1

optional in some configurations

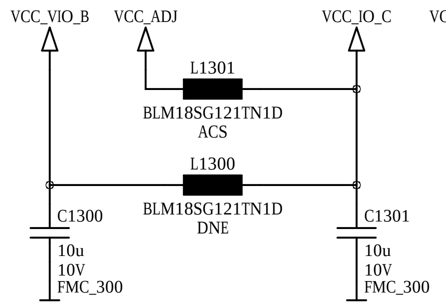


Signal Reordering

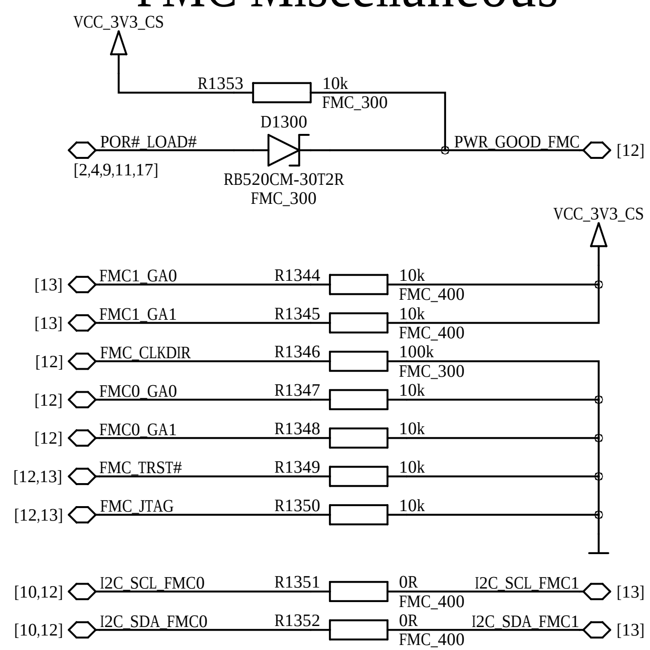




FMC Power



FMC Miscellaneous



FMC Infos

suggested connector for FMC HPC modules: ASP-134488-01

suggested connector for FMC LPC modules: ASP-134604-01

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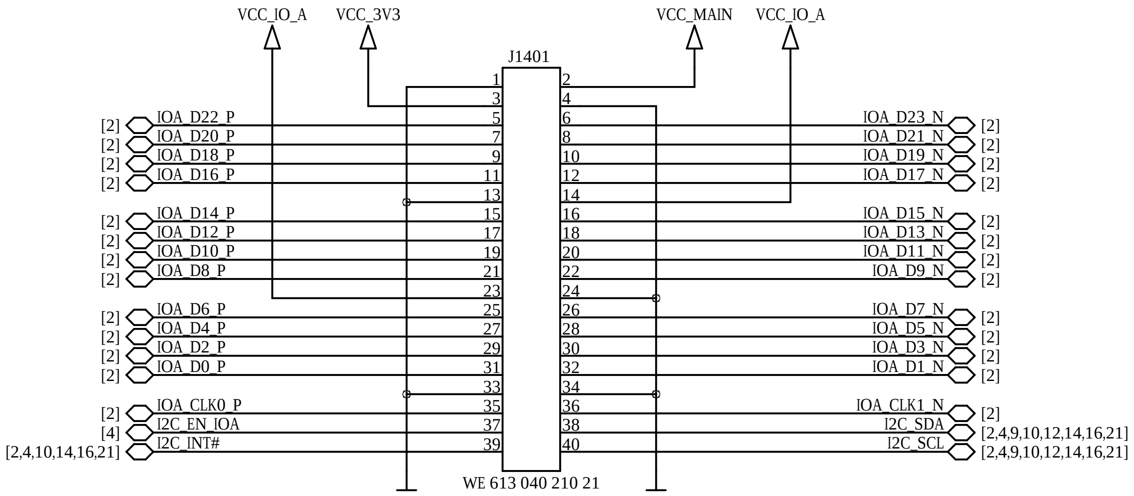
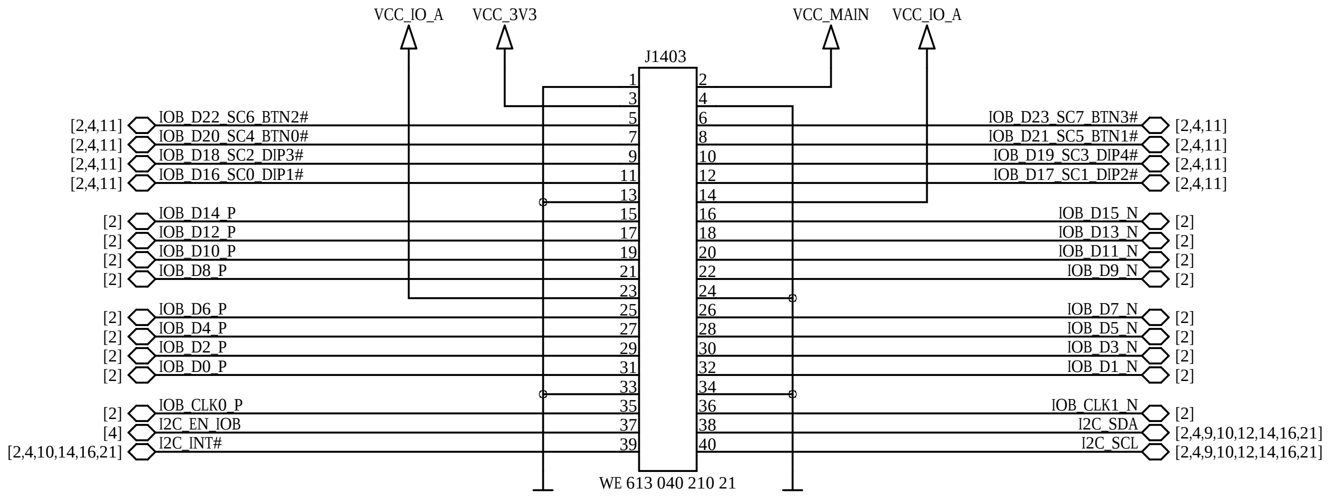
Anios IO Connector A

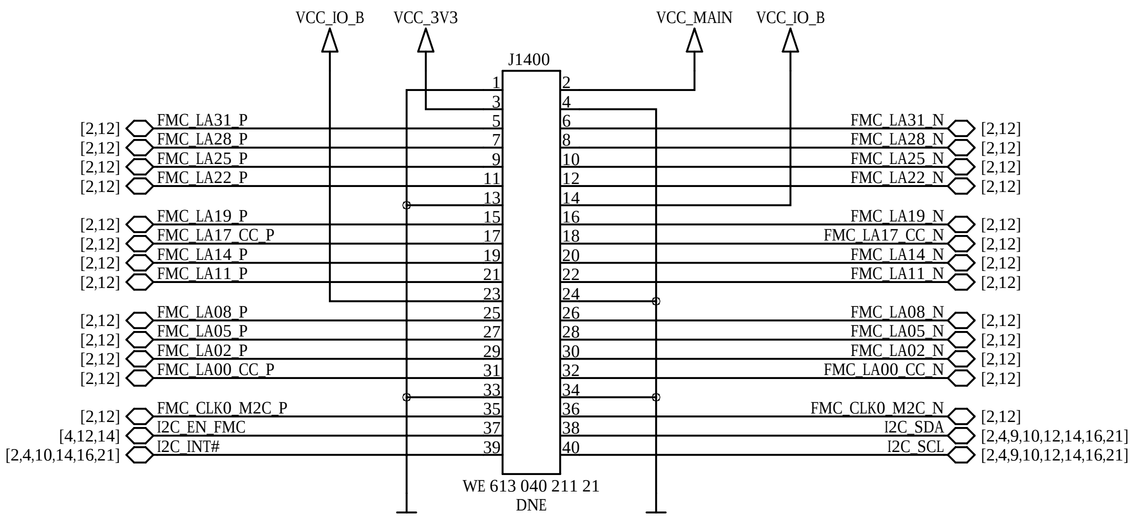


Anios IO Connector B



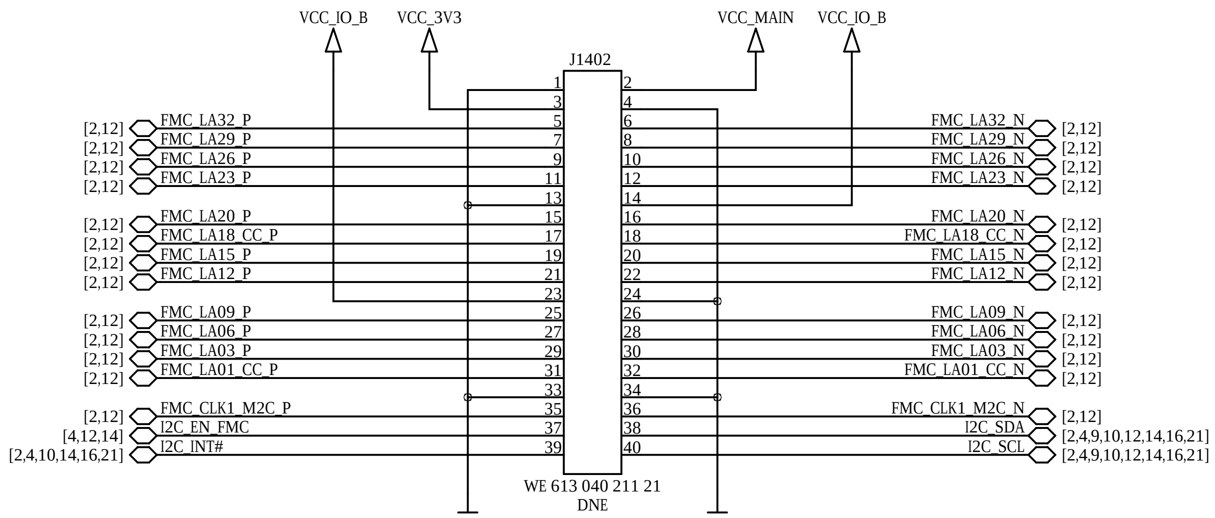
Anios IO Connector F

optional, shared with FMC connector



Anios IO Connector G

optional, shared with FMC connector



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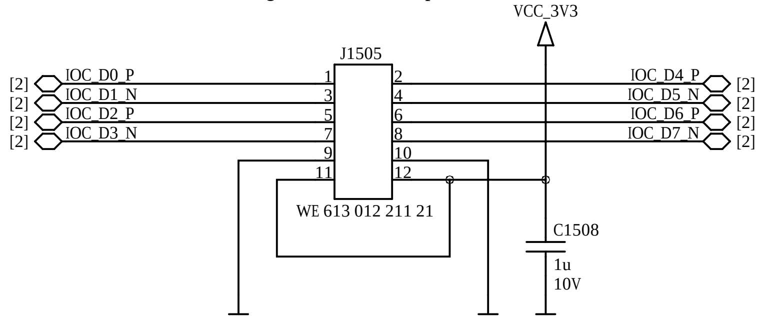
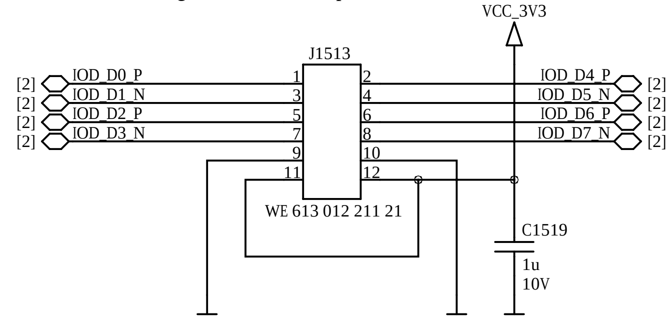
PMOD IO Connector C

This connector is Digilent Pmod™ Compatible when VCC_IO_B is 3.3V

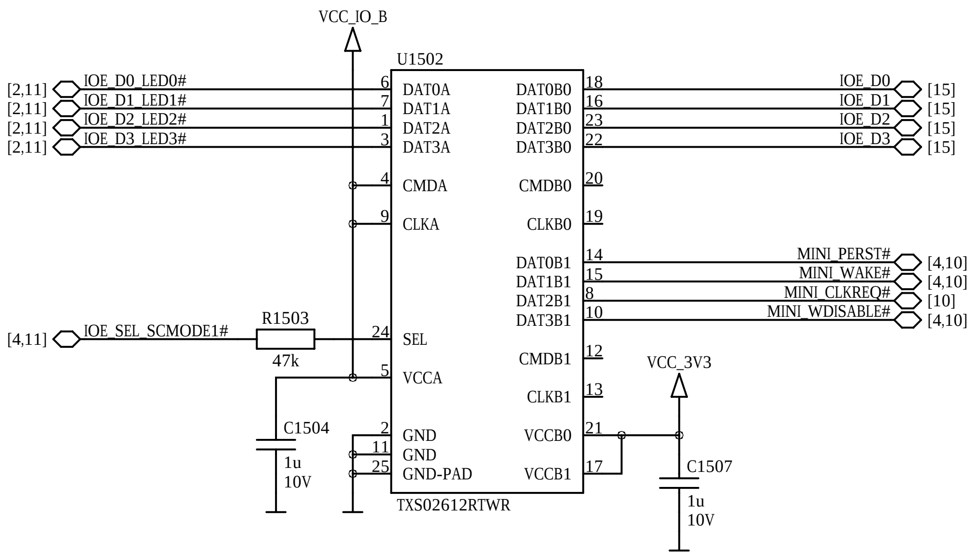


PMOD IO Connector D

This connector is Digilent Pmod™ Compatible when VCC_IO_B is 3.3V

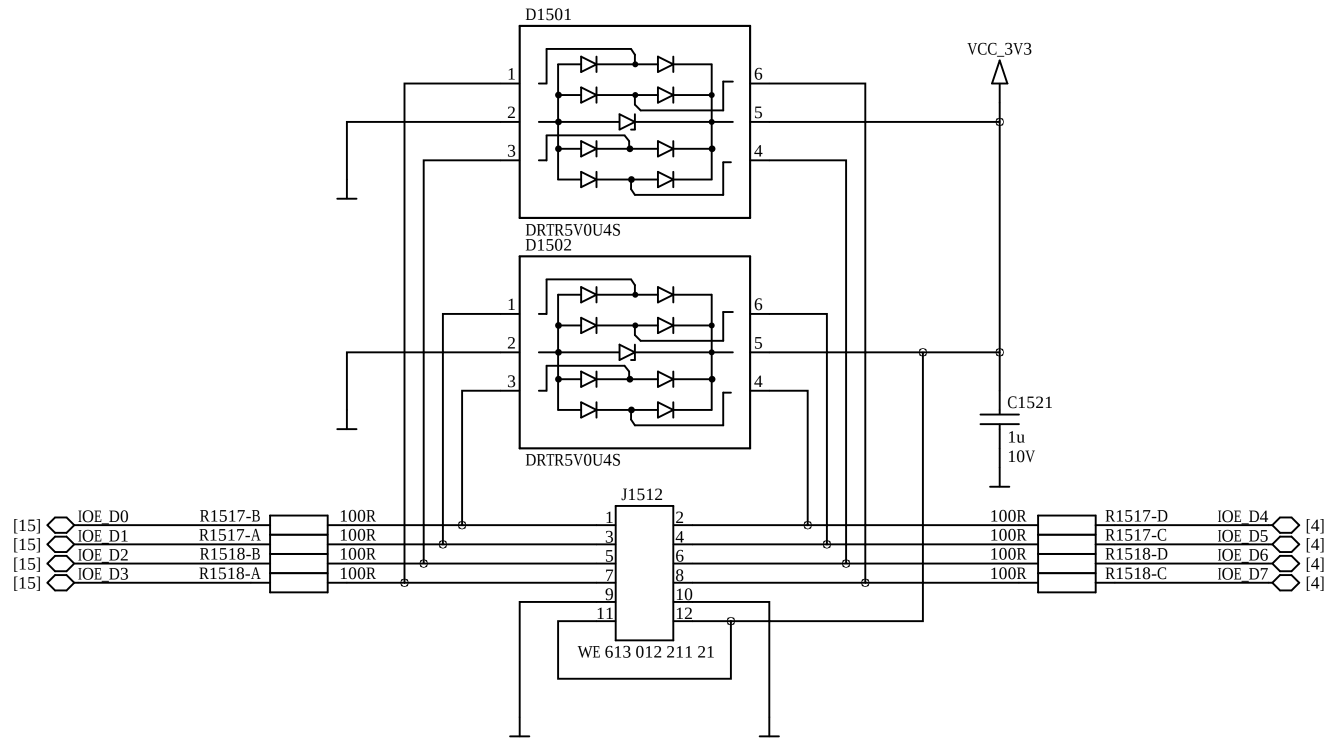


Level Shifter IO E



PMOD IO Connector E

This connector is Digilent Pmod™ Compatible



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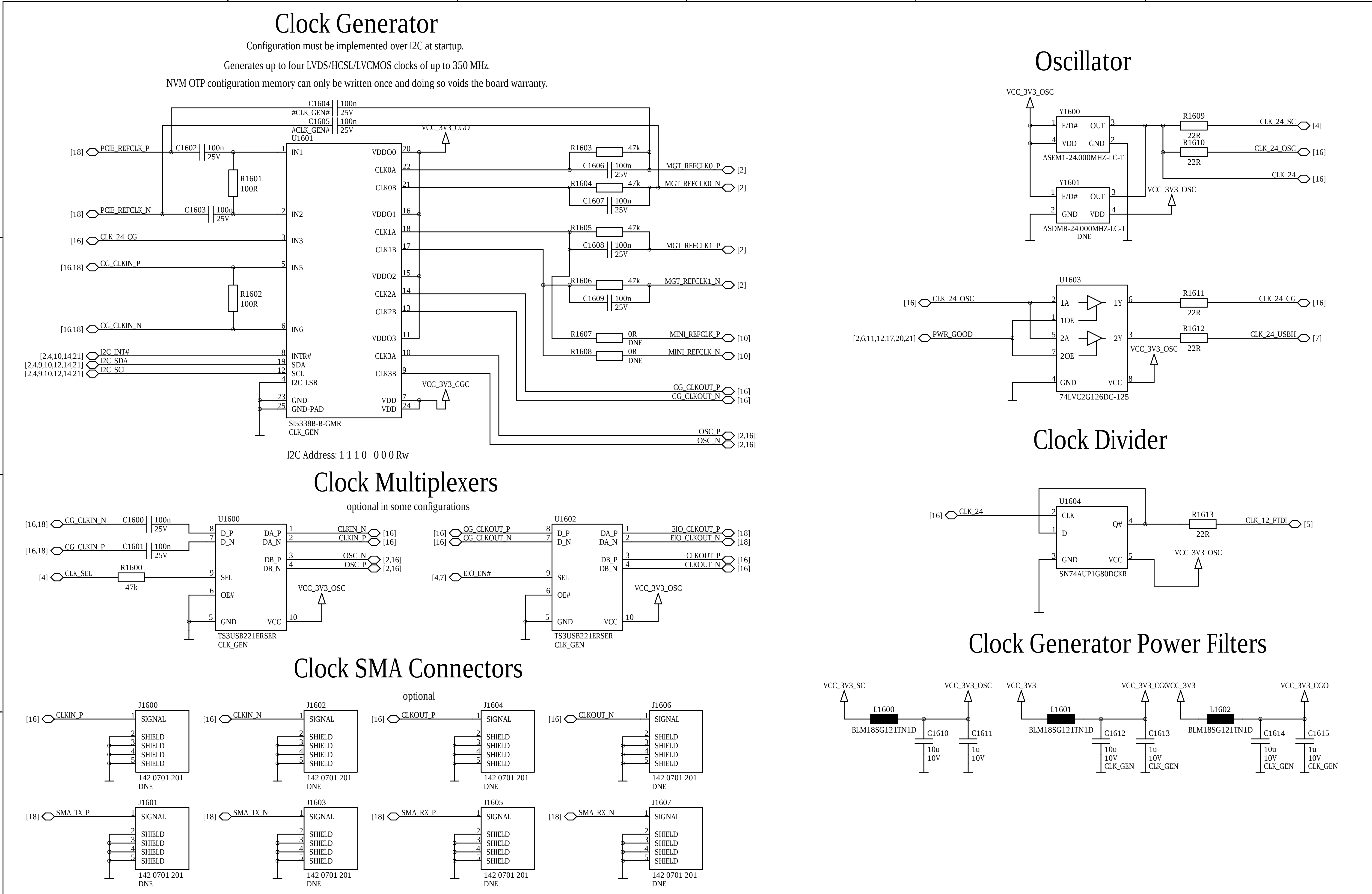

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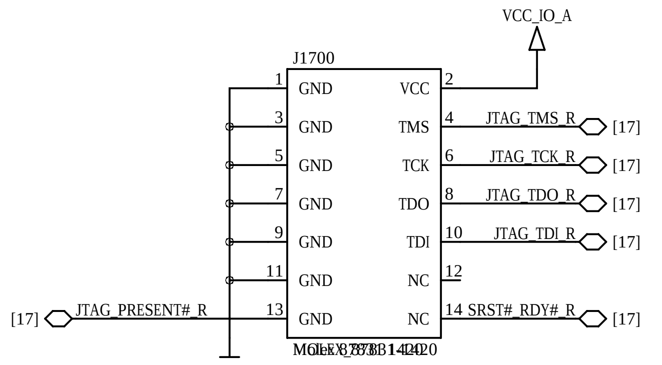
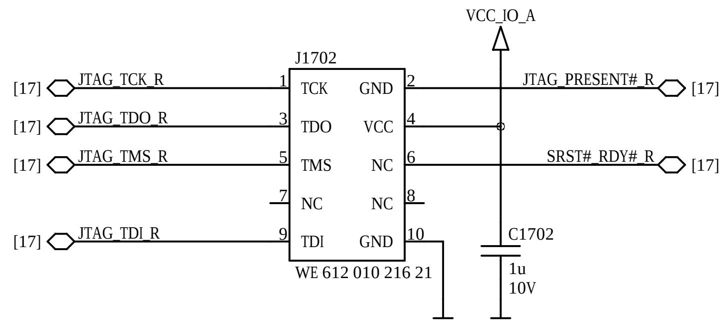
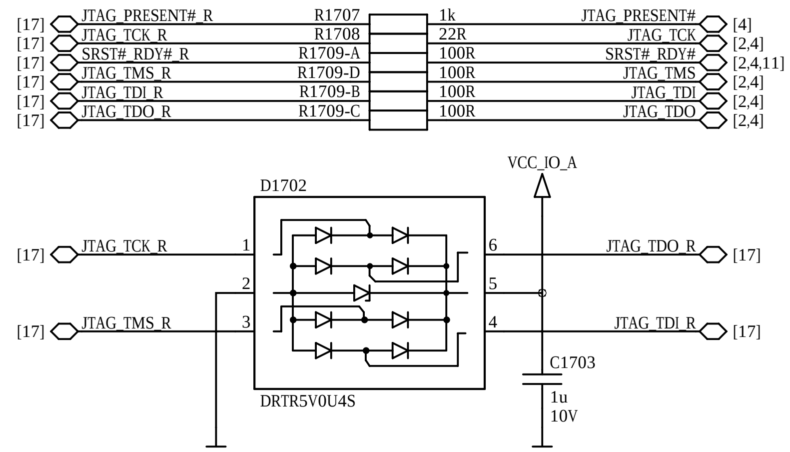
Xilinx JTAG Connector



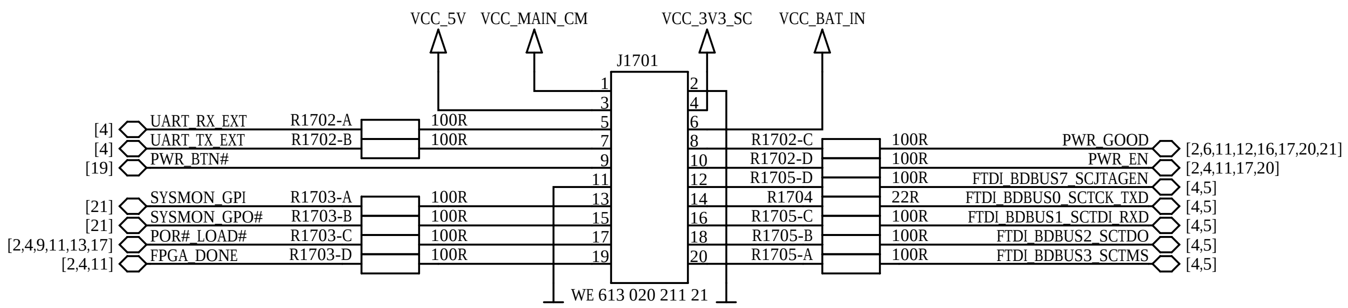
Altera JTAG Connector



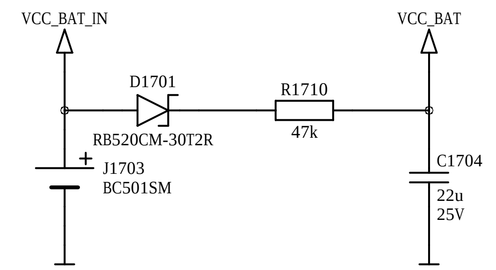
JTAG ESD Protection



Control Connector



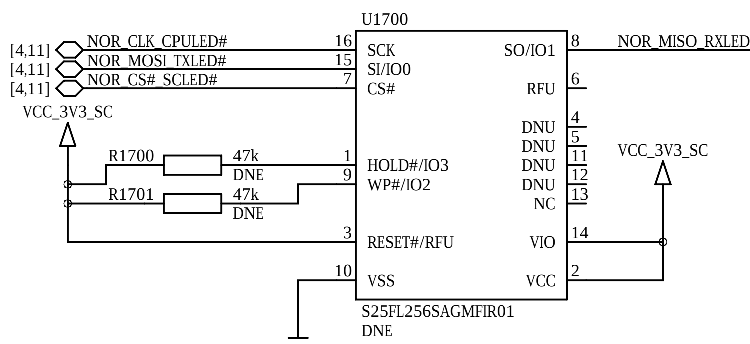
Battery Holder



SPI NOR Flash

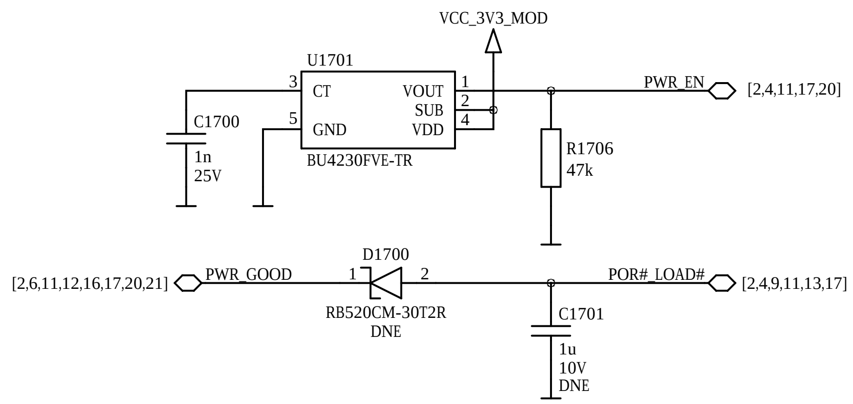
optional

pin compatible with Altera EPCS/EPCQ and Micron N25Q series



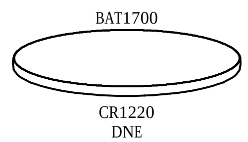
Reset Circuit

D1700, C1701 only needed when not using System Controller



Battery

optional



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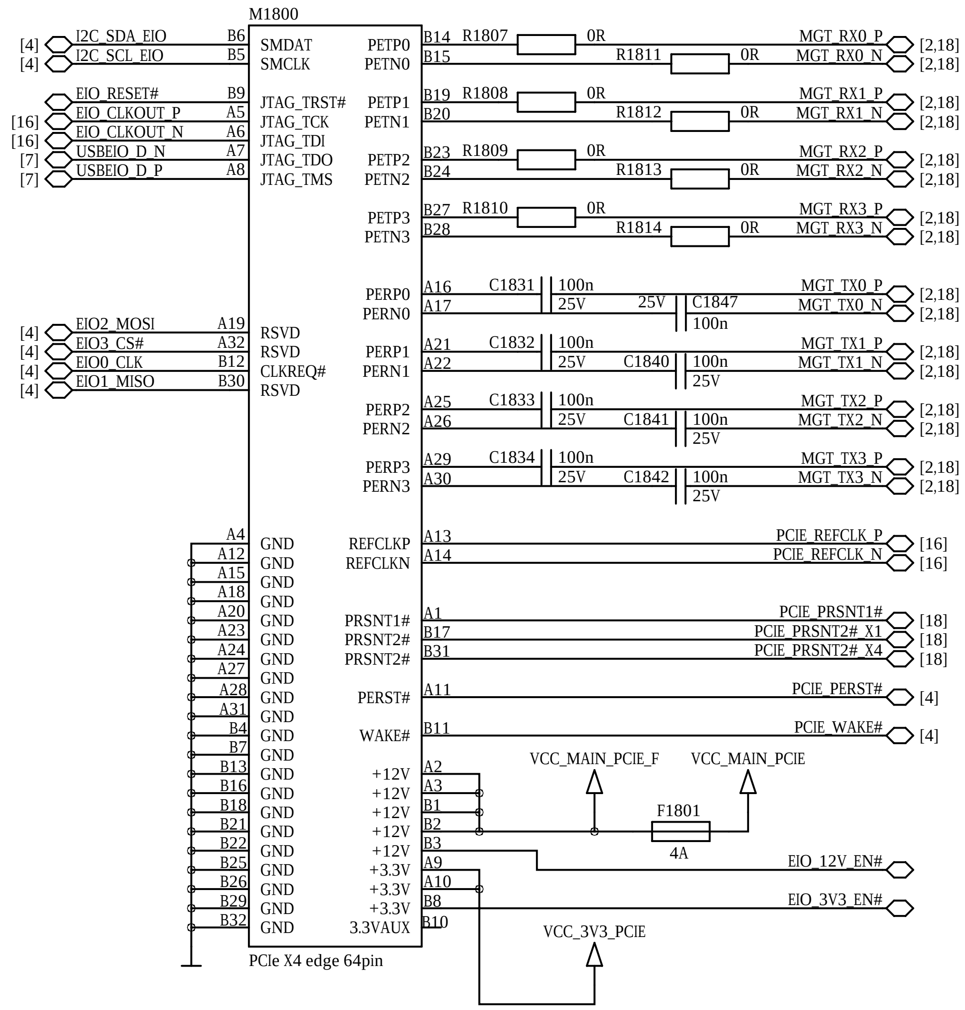
PCIe x4 Edge Connector



Edge IO Power

not included in user schematics

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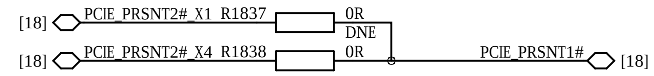
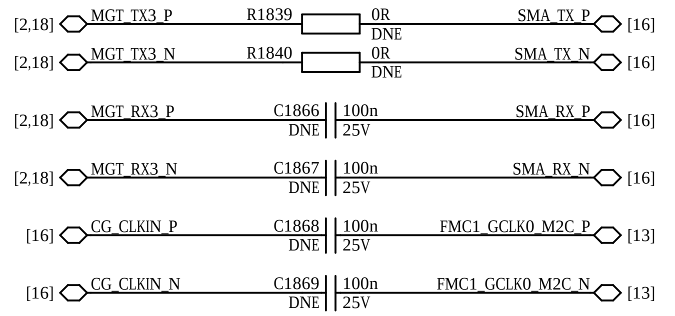
PCIe Present



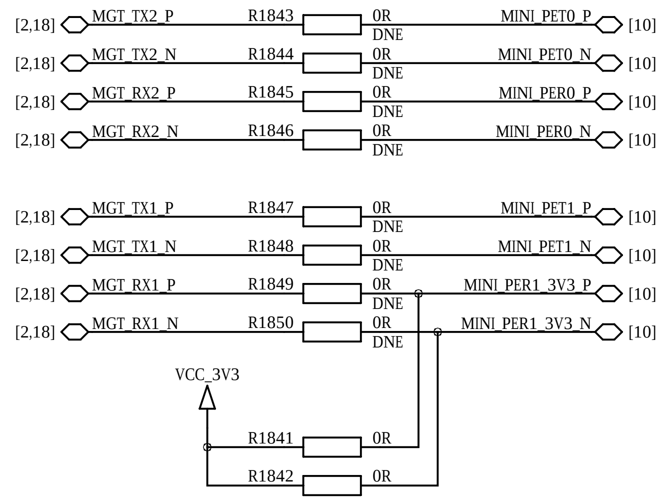
FMC MGT Signals

optional



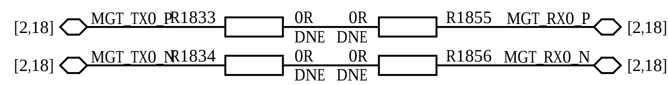
Mini PCIe / Mini SATA Card Signals

optional



MGT Loopback

optional



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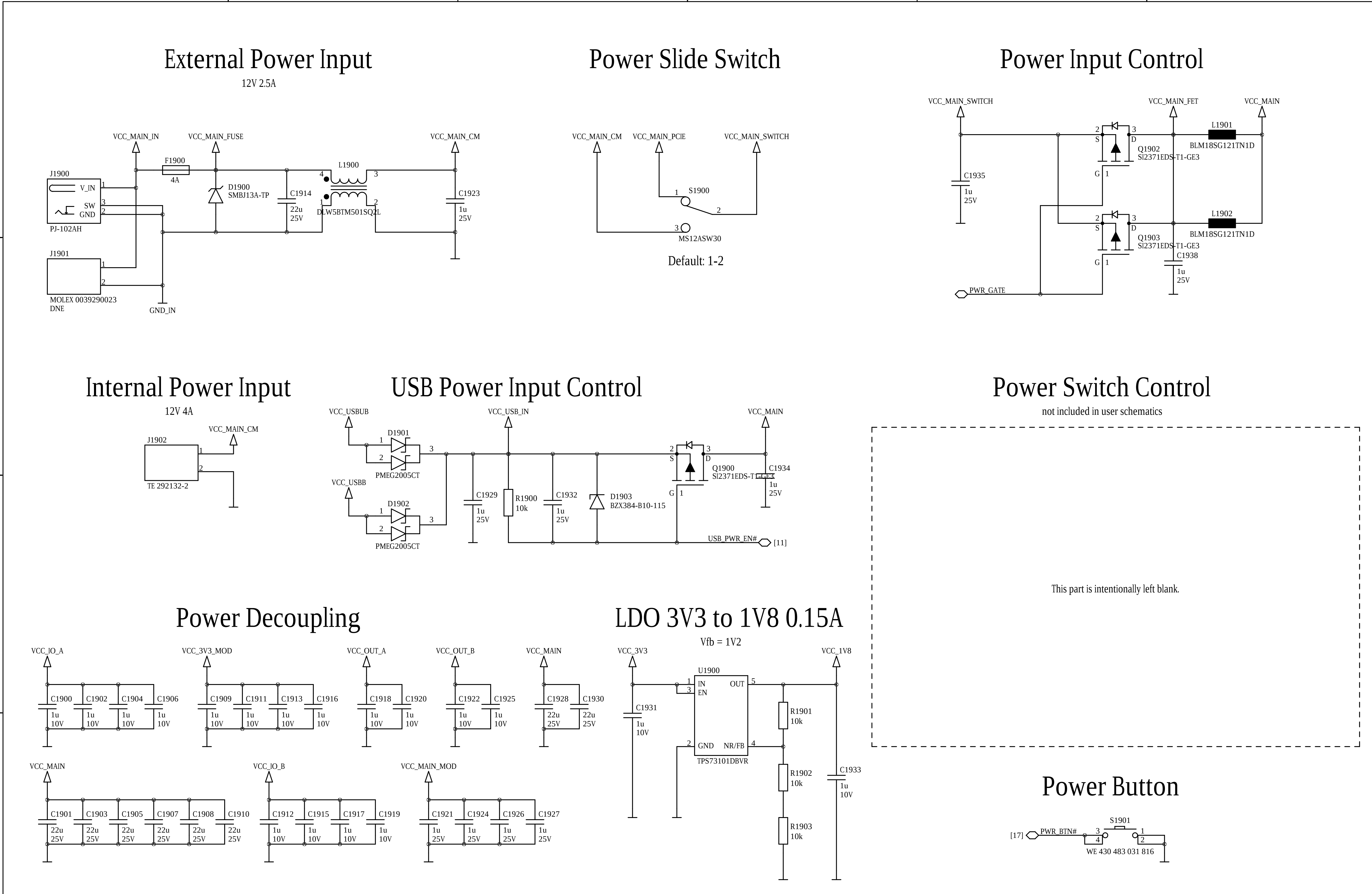

ENCLUSTRA	Copyright	© 2020 by Enclustra GmbH	Sheet Name	19_POWER_INPUT_CONTROL	Customer No	0000	Revision	R4.6	DNE = do not equip	Confidential
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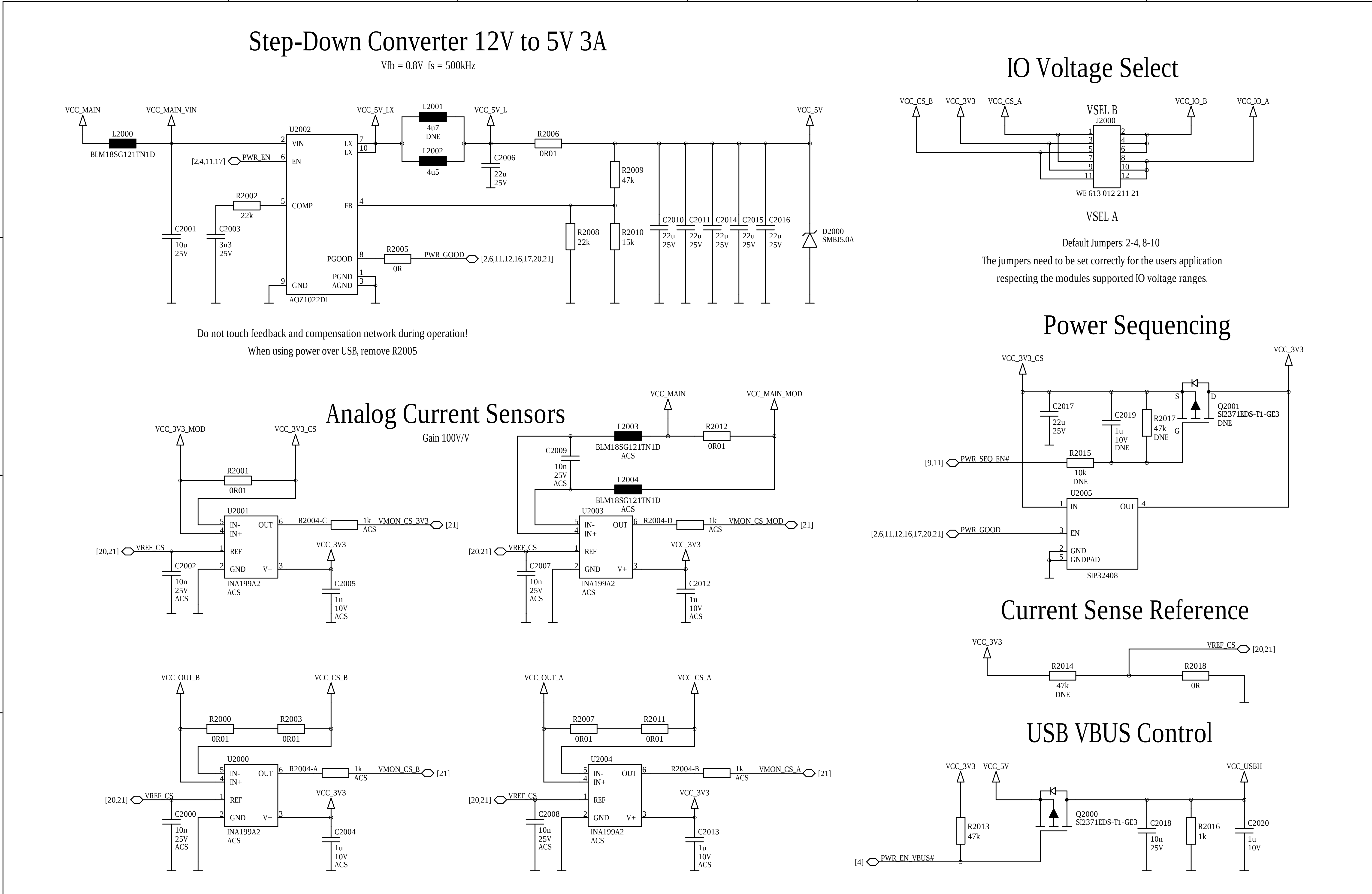

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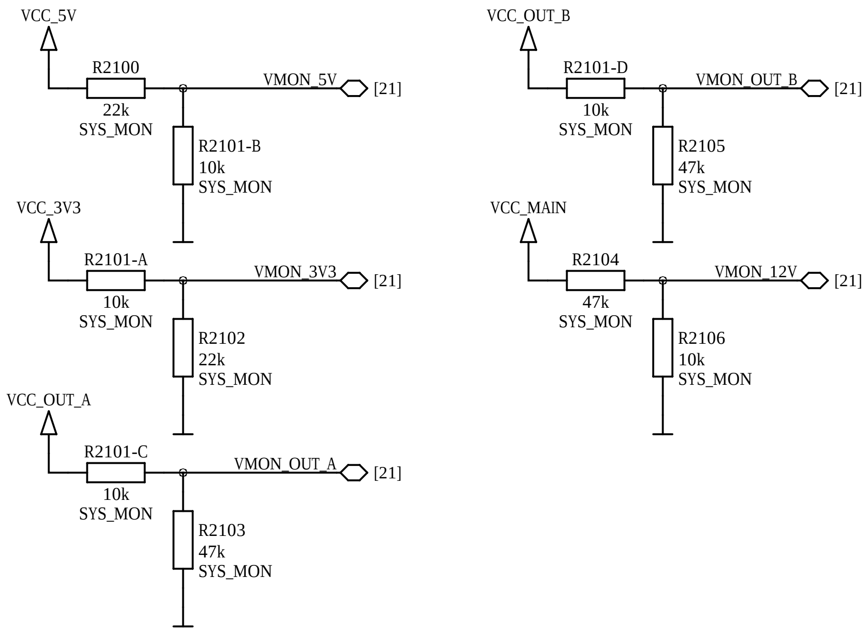
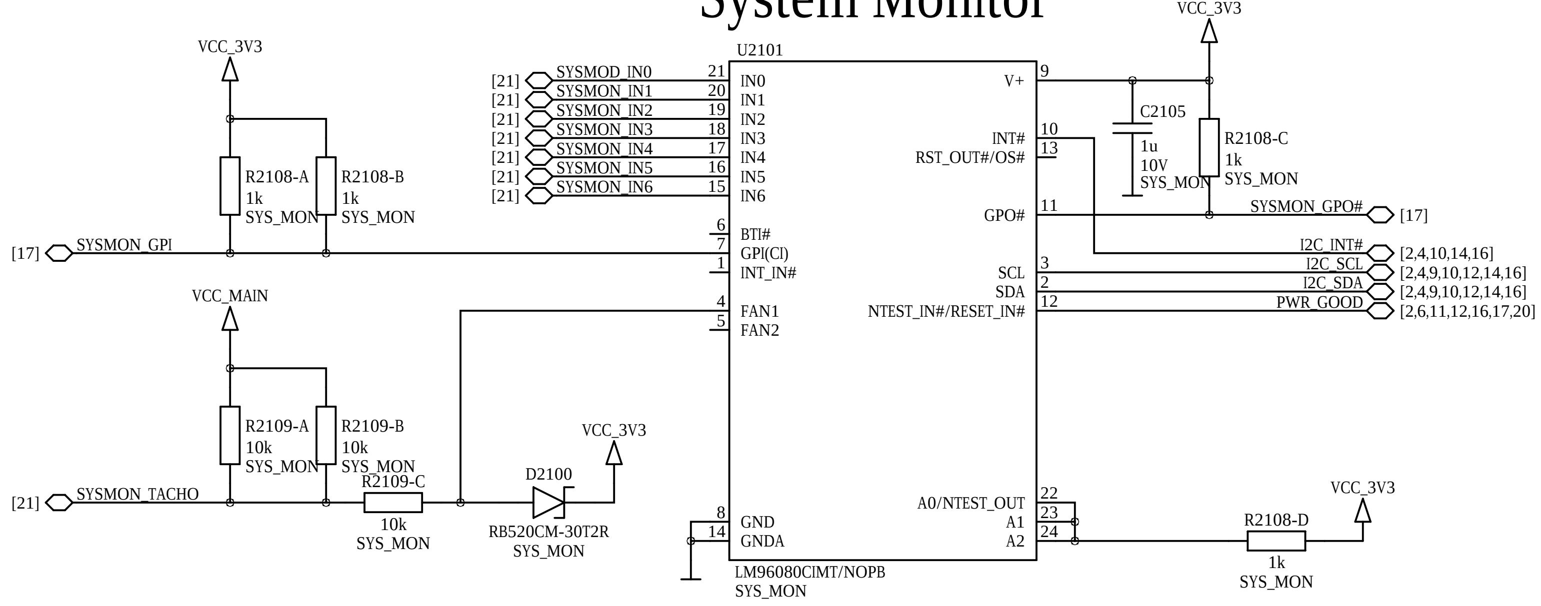
System Monitor Voltage Sense

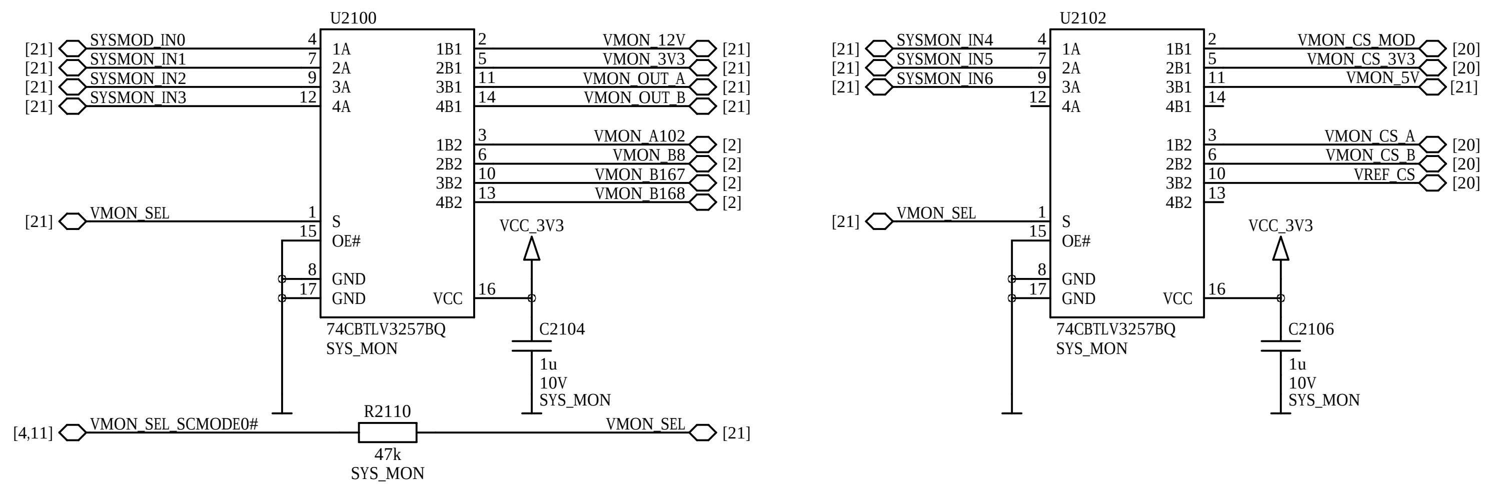


System Monitor

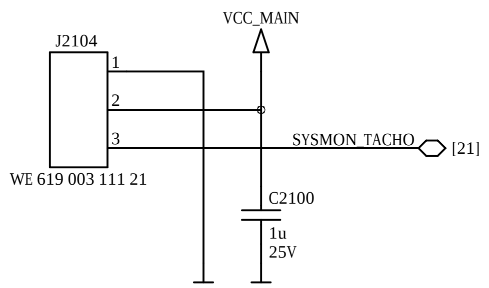


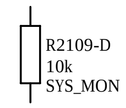
I2C Address: 0 1 0 1 1 1 1 Rw

System Monitor Analog Multiplexers



Fan Connector





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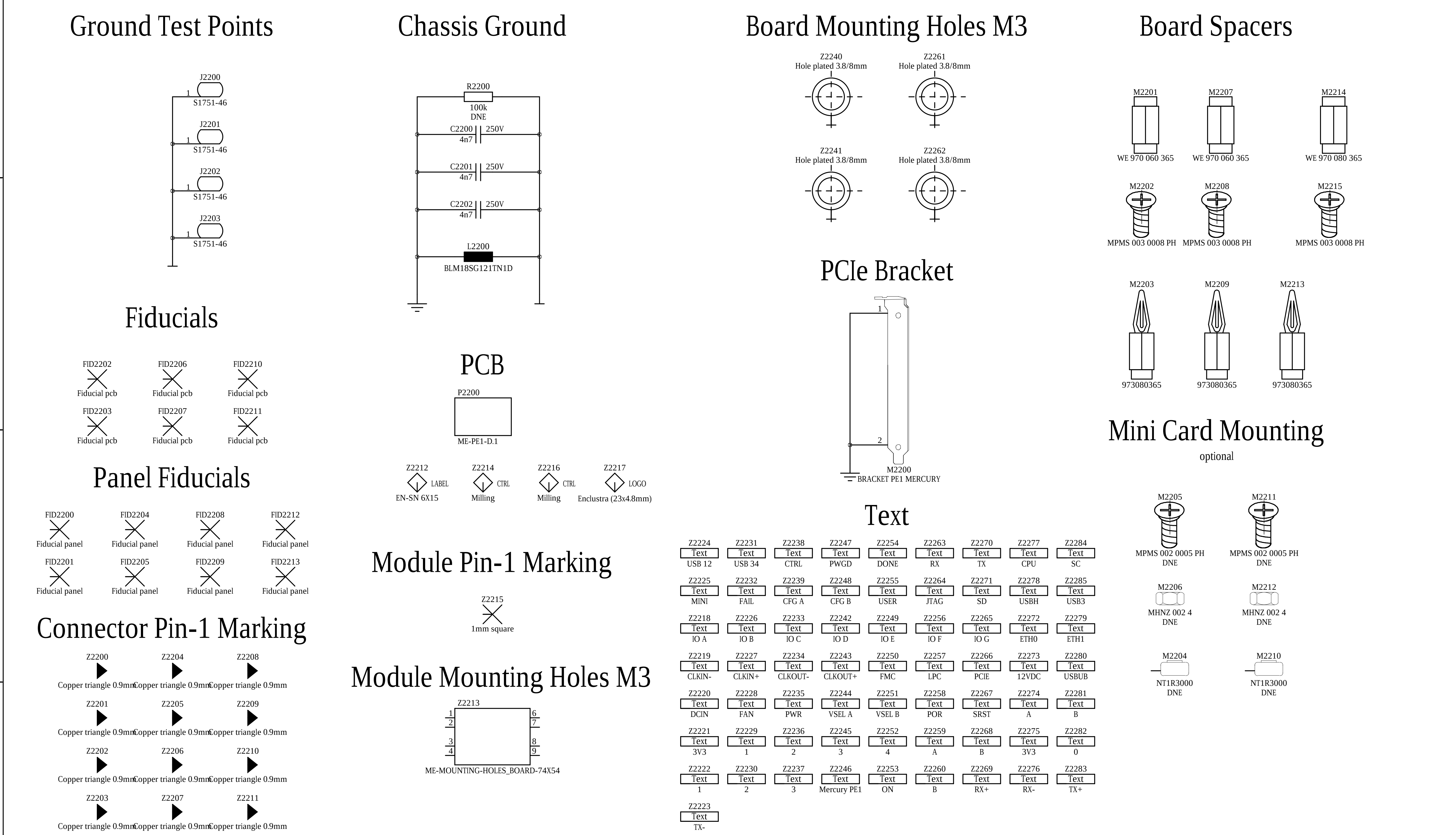

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Component Variants

SYS_MON	FMC	EMMC	SIMCARD	MOD_CONN	ACS	CLK_GEN	FMC_300	FMC_400
page 21	Part J1200A ASP-134603-01 EN100364 FMC_HPC_LPC_GENERIC Part J1200B ASP-134486-01 EN101301 FMC_HPC_LPC_GENERIC	Part U900A:1 EMMC04G-M627-X03U EN102717 Part U900A:2 H26M31001HPR EN101269	page 10	page 3	pages 6, 7, 13 and 20	page 16	page 13	page 13

Assembly Variants

Assembly Variant	SYS_MON	FMC	EMMC	SIMCARD	MOD_CONN	ACS	CLK_GEN	FMC_300	FMC_400
Variant EN103080:0 ME-PE1-200-C-R4.6	Option EN103080:1 SYS_MON DNE	Option EN103080:2 FMC_FEMALE EN100364	Option EN103080:3 EMMC DNE	Option EN103080:4 SIMCARD DNE	Option EN103080:5 MOD_CONN DNE	Option EN103080:6 ACS DNE	Option EN103080:7 CLK_GEN DNE	Option EN103080:8 FMC_300 DNE	Option EN103080:9 FMC_400 DNE
Variant EN103081:0 ME-PE1-300-W-R4.6	Option EN103081:1 SYS_MON	Option EN103081:2 FMC_FEMALE EN101301	Option EN103081:3 EMMC EN101269	Option EN103081:4 SIMCARD	Option EN103081:5 MOD_CONN	Option EN103081:6 ACS	Option EN103081:7 CLK_GEN	Option EN103081:8 FMC_300	Option EN103081:9 FMC_400 DNE
Variant EN103082:0 ME-PE1-400-W-R4.6	Option EN103082:1 SYS_MON	Option EN103082:2 FMC_FEMALE EN100364	Option EN103082:3 EMMC EN101269	Option EN103082:4 SIMCARD	Option EN103082:5 MOD_CONN	Option EN103082:6 ACS	Option EN103082:7 CLK_GEN	Option EN103082:8 FMC_300 DNE	Option EN103082:9 FMC_400